

Numerical & mathematical skills for neuroscience

PS5210 2025-26

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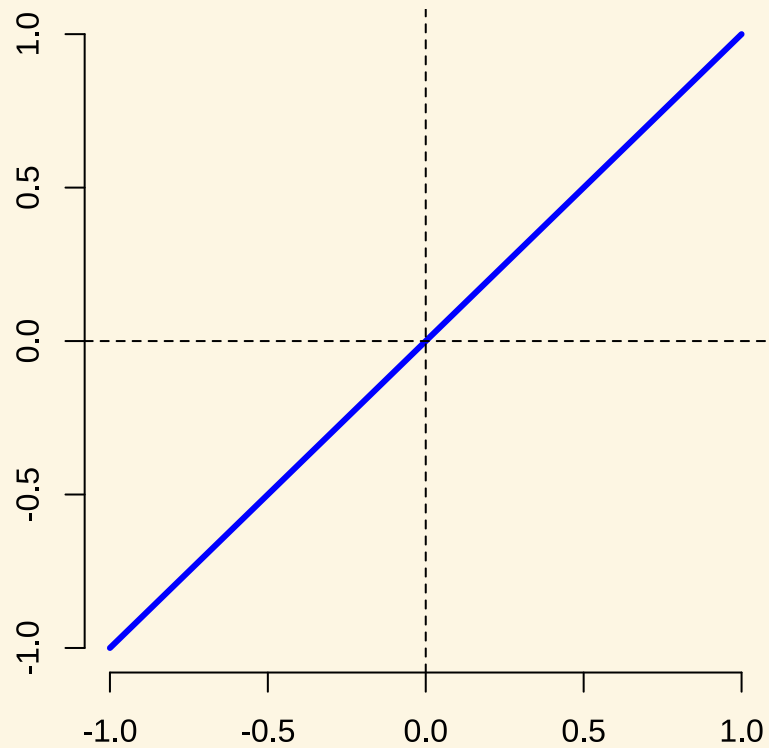
Part 1

Program for today

- **Part 1** Mathematical functions, some calculus and Fourier transformation
- **Part 2** Introduction to linear algebra

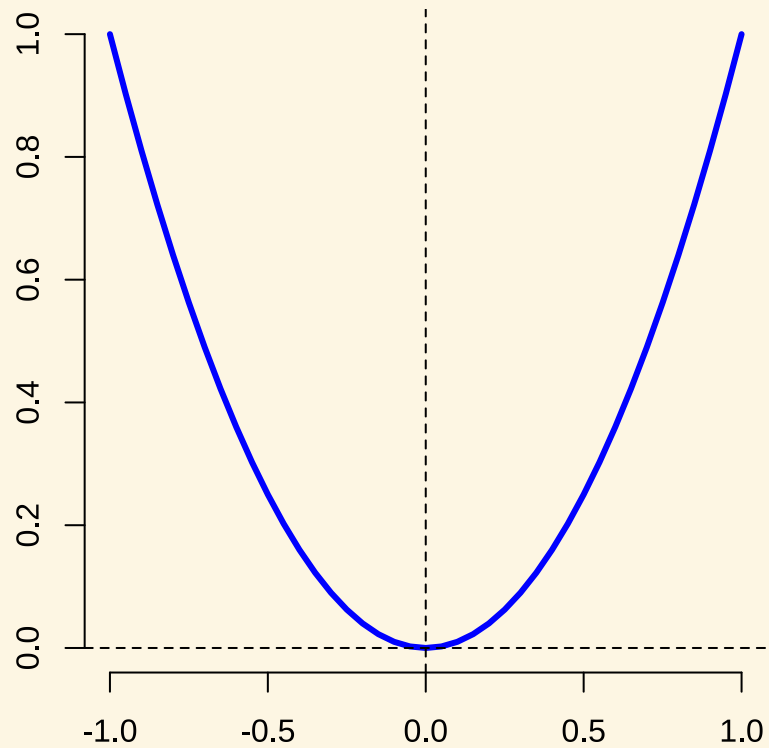
Polynomial functions

$$y = x$$



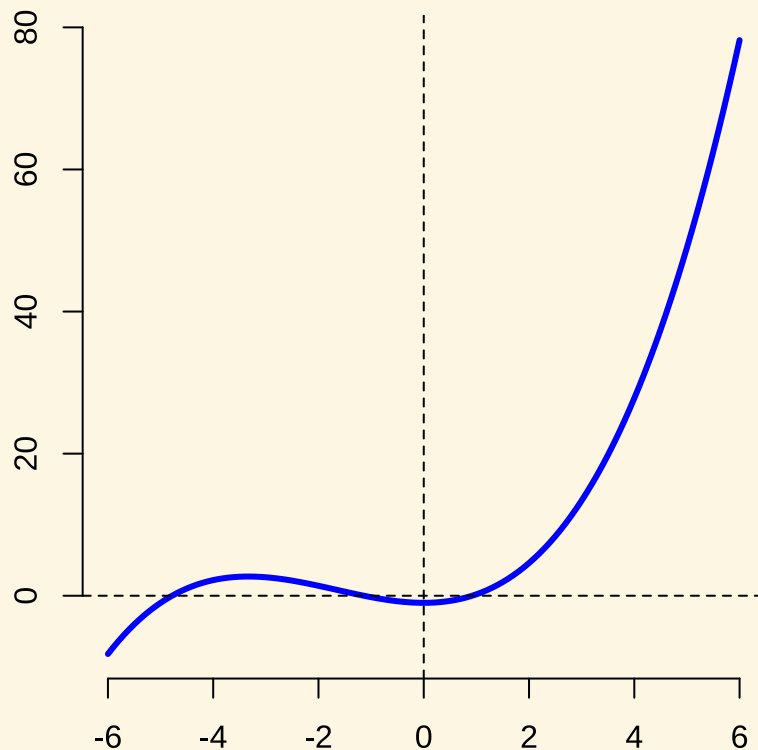
Polynomial functions

$$y = x^2$$



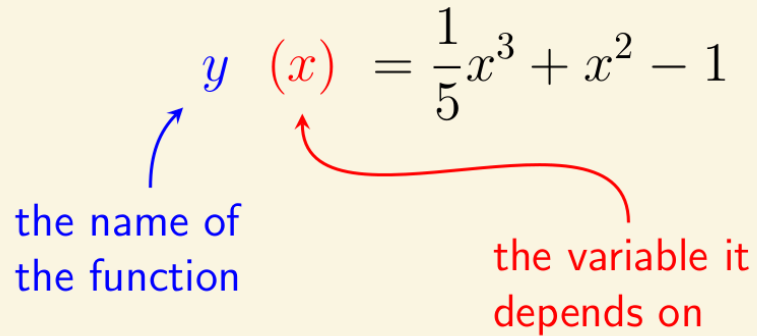
Polynomial functions

$$y = \frac{1}{5}x^3 + x^2 - 1$$



Function notation

To show that y is a function of x we write



The diagram shows the function notation $y(x) = \frac{1}{5}x^3 + x^2 - 1$. A blue arrow points from the text "the name of the function" to the y . A red arrow points from the text "the variable it depends on" to the (x) .

$$y(x) = \frac{1}{5}x^3 + x^2 - 1$$

the name of the function

the variable it depends on

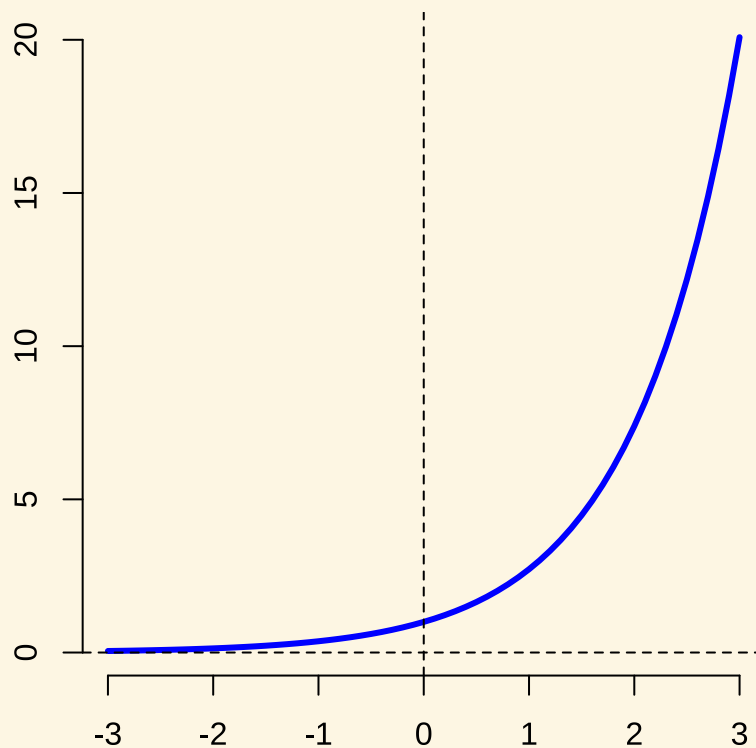
if a function depends on multiple variables we write

$$y(x, t) = x^2 + t + xt$$

Exponential function

$$y = \exp(x) = e^x$$

where e is the Euler number $e = 2.7182818284590452353 \dots$

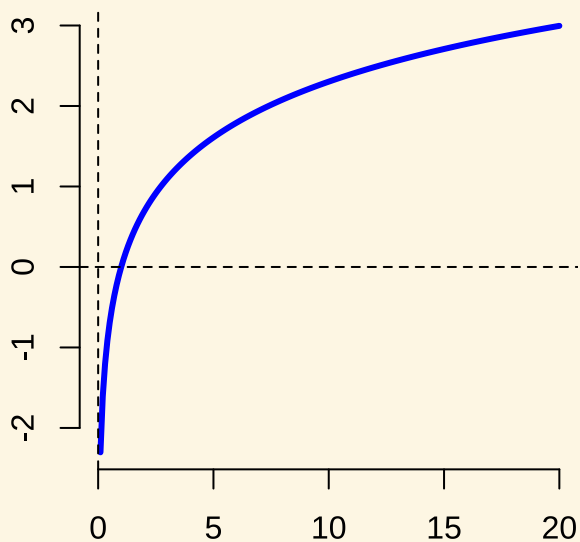


Logarithm function

Logarithmic function with base e

$$y = \log_e(x) = \ln(x)$$

It's the inverse of the exponential function: if $y = e^x$, then $x = \log_e(y)$



Logarithmic Arithmetic

Multiplying two exponential leads to addition:

$$b^{x_1} \times b^{x_2} = b^{x_1+x_2}$$

Conversely, the sum of two logarithms can be expressed as the logarithm of their product

$$\log(x_1 \times x_2) = \log(x_1) + \log(x_2)$$

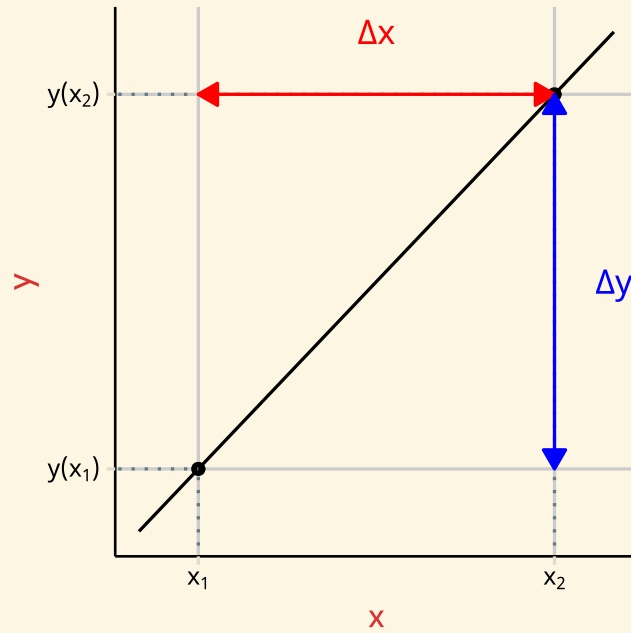
This property comes hand when we are calculating the *joint probability* of many independent events. This is computed as the product of individual probabilities, and it is often convenient to turn this into a sum using logarithms.

Differentiation

- How does a function y vary with x ?
- that is, given a change in x of Δx , what is Δy , the corresponding change in y ?

Differentiation

For a straight line:

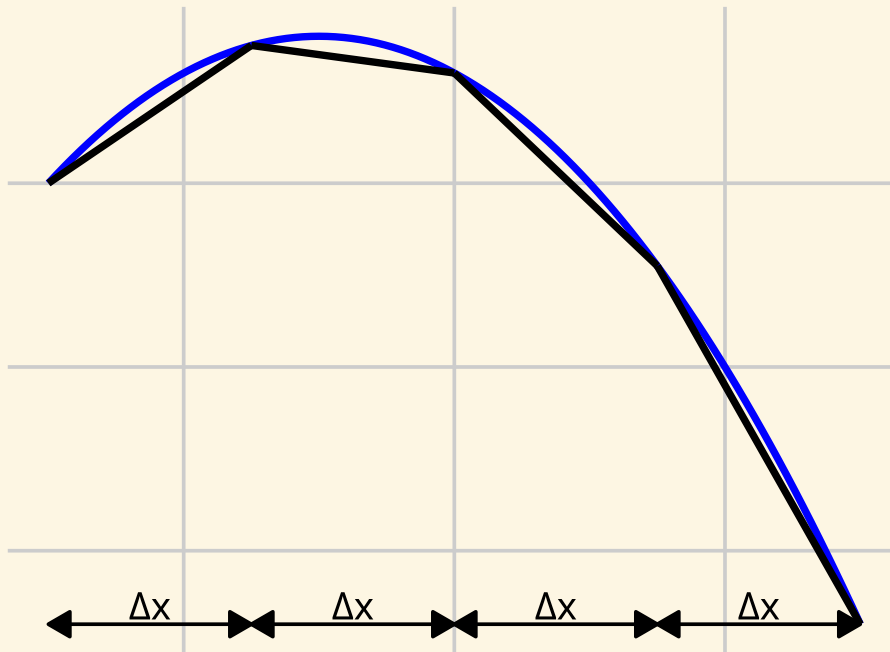


The *gradient* is:

$$\frac{\Delta y}{\Delta x} = \frac{y(x_2) - y(x_1)}{x_2 - x_1}$$

Differentiation

A curve can be approximated with a series of lines each of length Δx

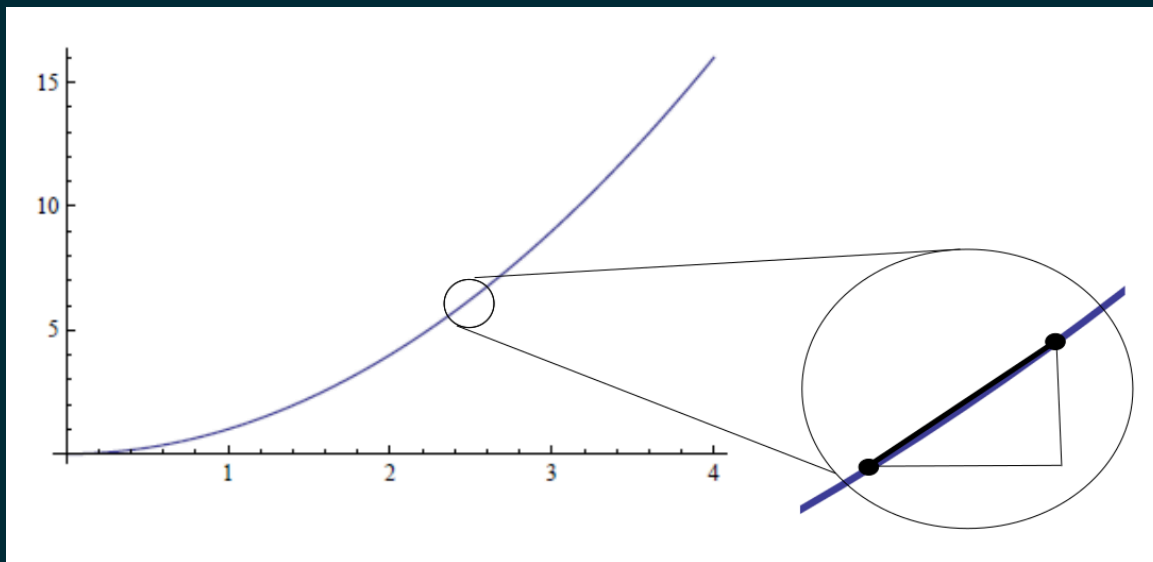


The shorter the lines, the more accurate the approximation.

It becomes *precise* as $\Delta x \rightarrow 0$

Differentiation

Differentiating the function $y(x) = x^2$



$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{y(x + \Delta x) - y(x)}{(x + \Delta x) - x}$$

Differentiation

Differentiating the function $y(x) = x^2$

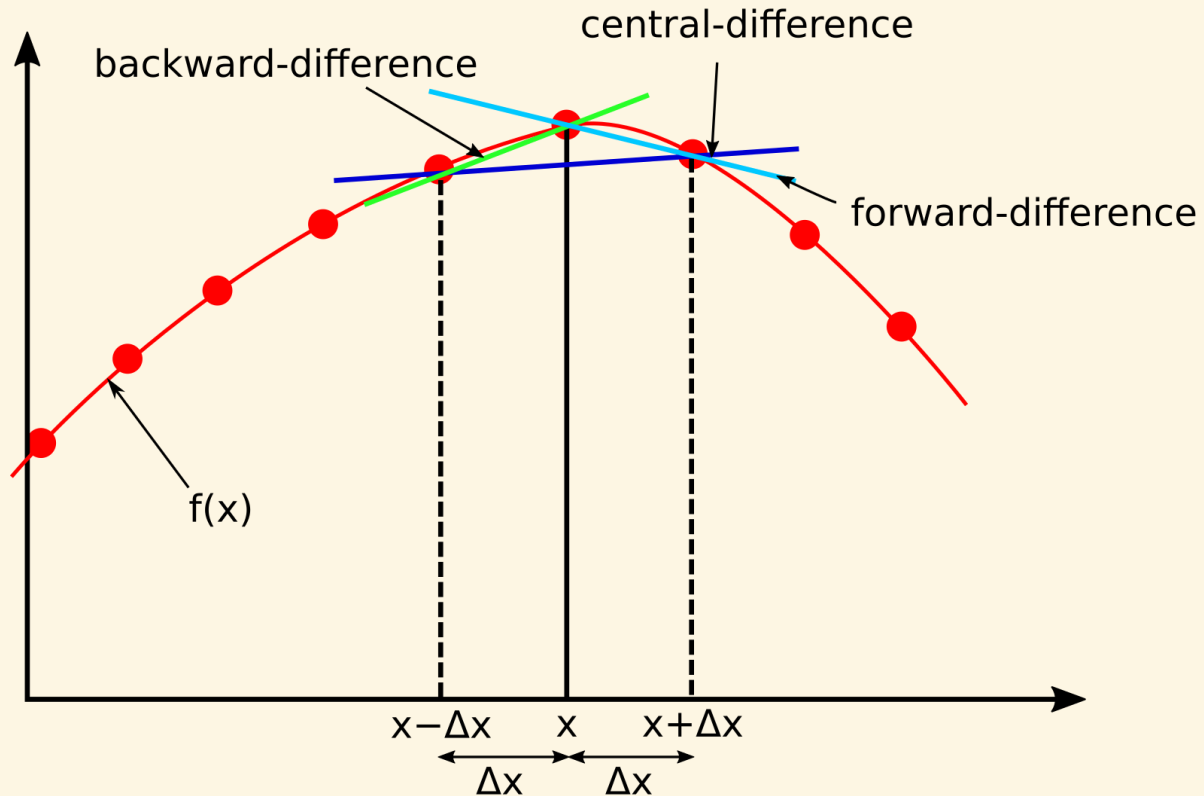
$$\begin{aligned}\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{(x+h) - x} \\&= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\&= \lim_{h \rightarrow 0} 2x + h \\&= 2x\end{aligned}$$

Differentiation

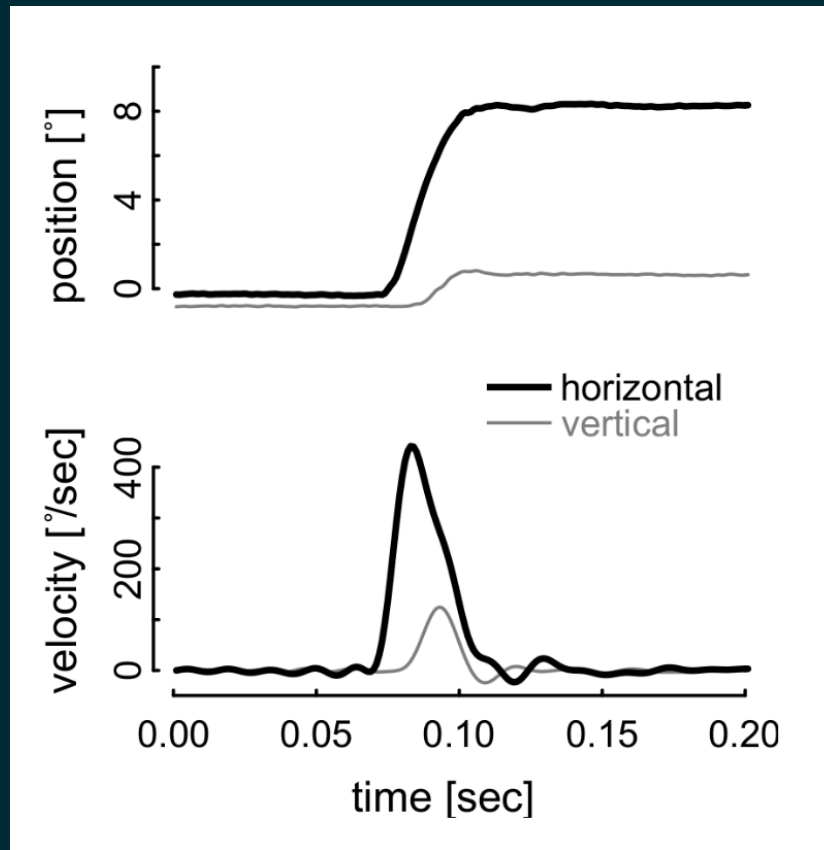
$y(x)$	$\frac{dy}{dx}$
x	1
x^2	$2x$
x^3	$3x^2$
$\vdots$	$\vdots$
x^n	nx^{n-1}
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
e^x	e^x

Numerical differentiation

For digital signals that are described by a discrete set of points or sample we can estimate the derivative using a *finite difference* approach.



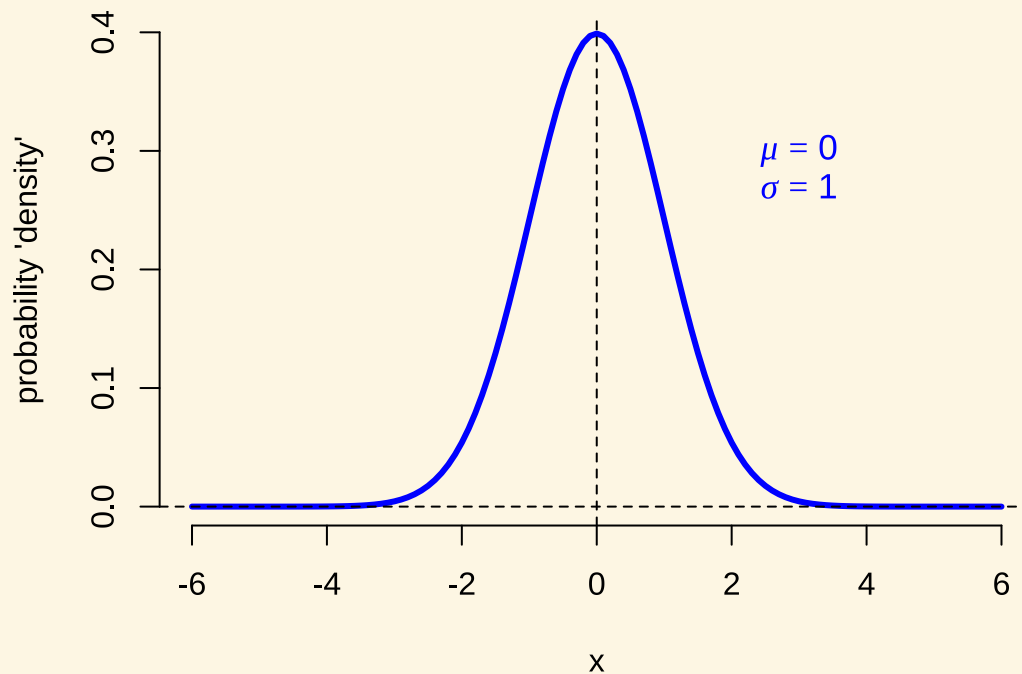
Example: differentiation gaze position to compute gaze velocity



Probability *density* functions

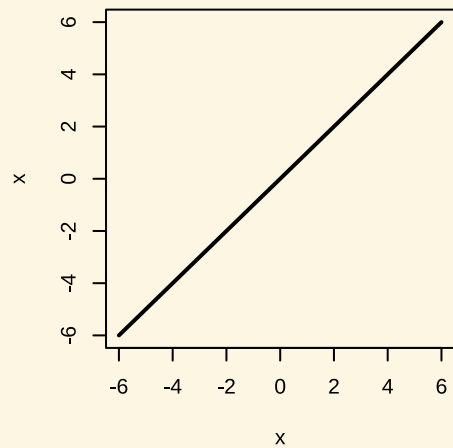
Normal or Gaussian distribution:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

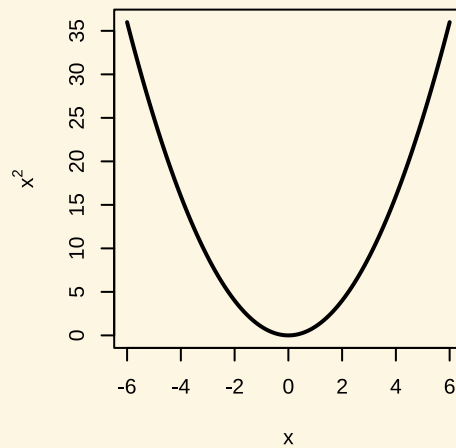


Understanding the equation of the normal distribution

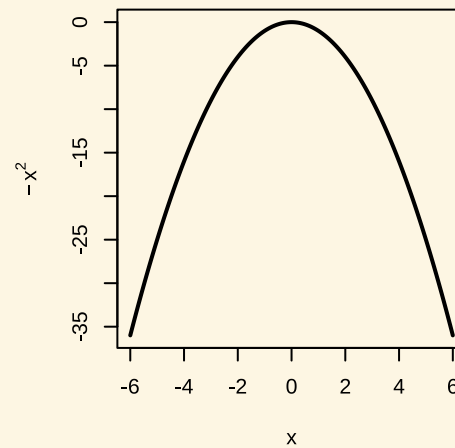
(a)



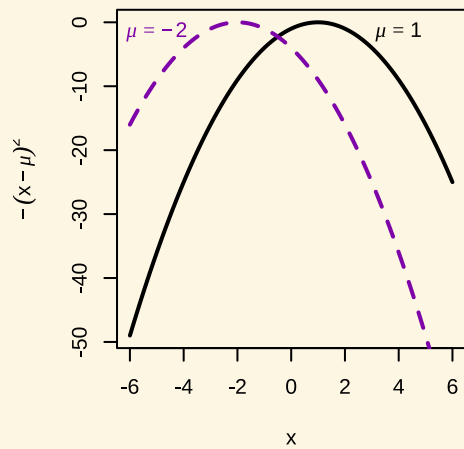
(b)



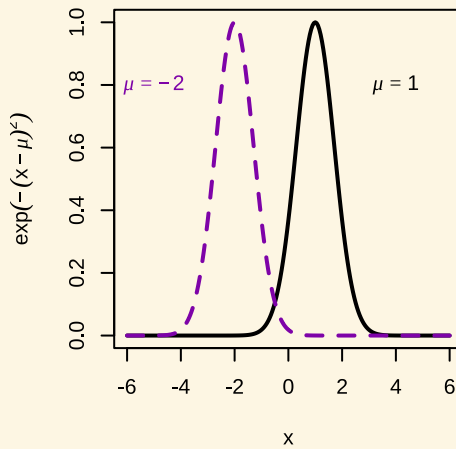
(c)



(d)



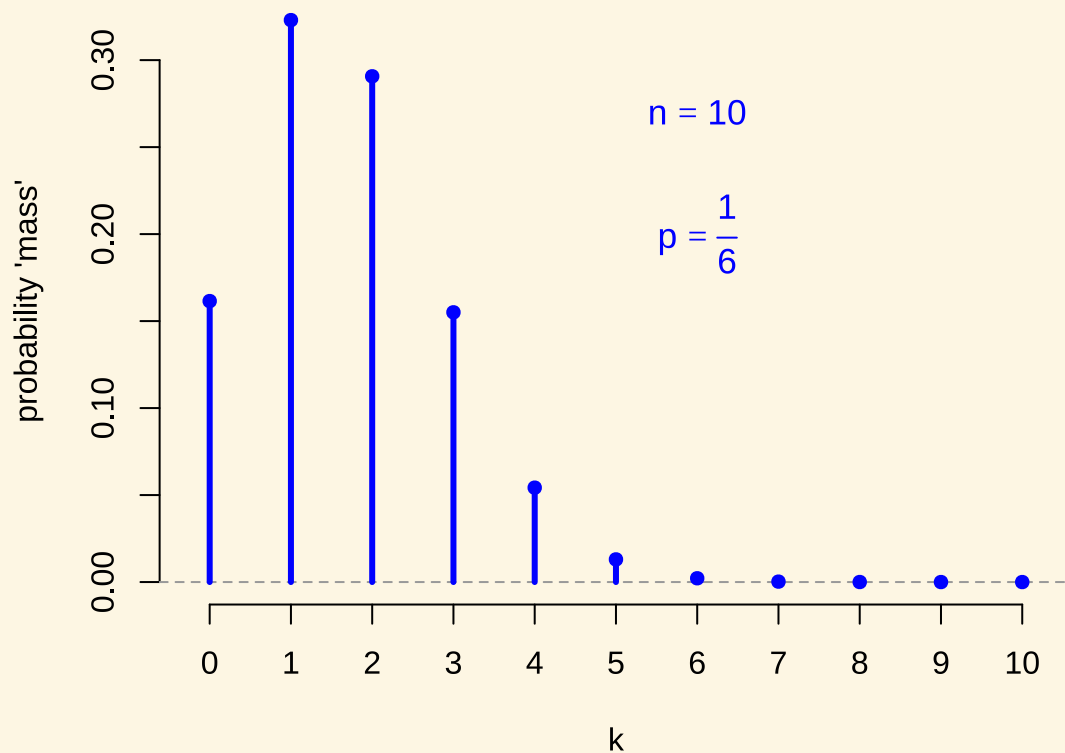
(e)



Probability *mass* functions

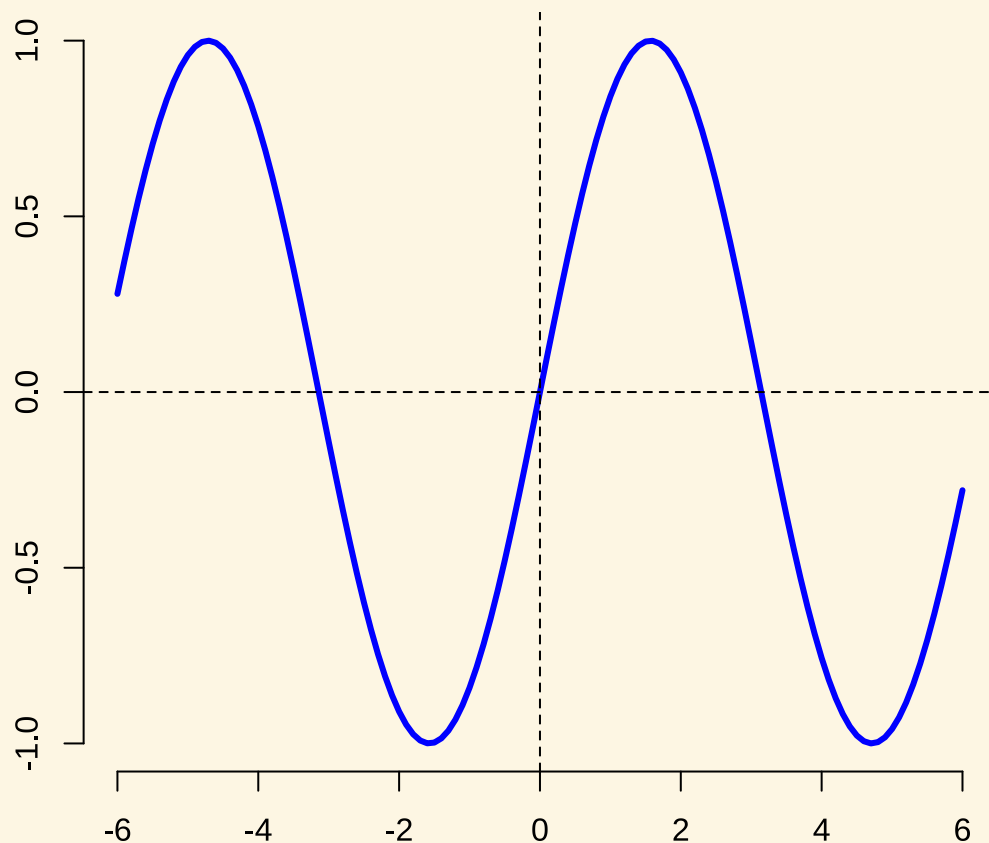
Binomial distribution:

$$\Pr(x) = \binom{n}{k} p^k (1 - p)^{n-k}$$

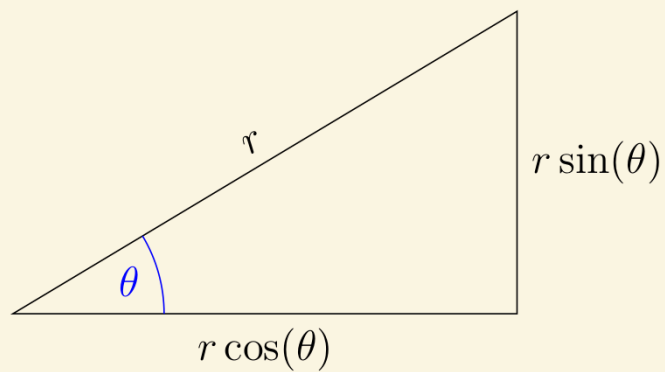


Trigonometric functions

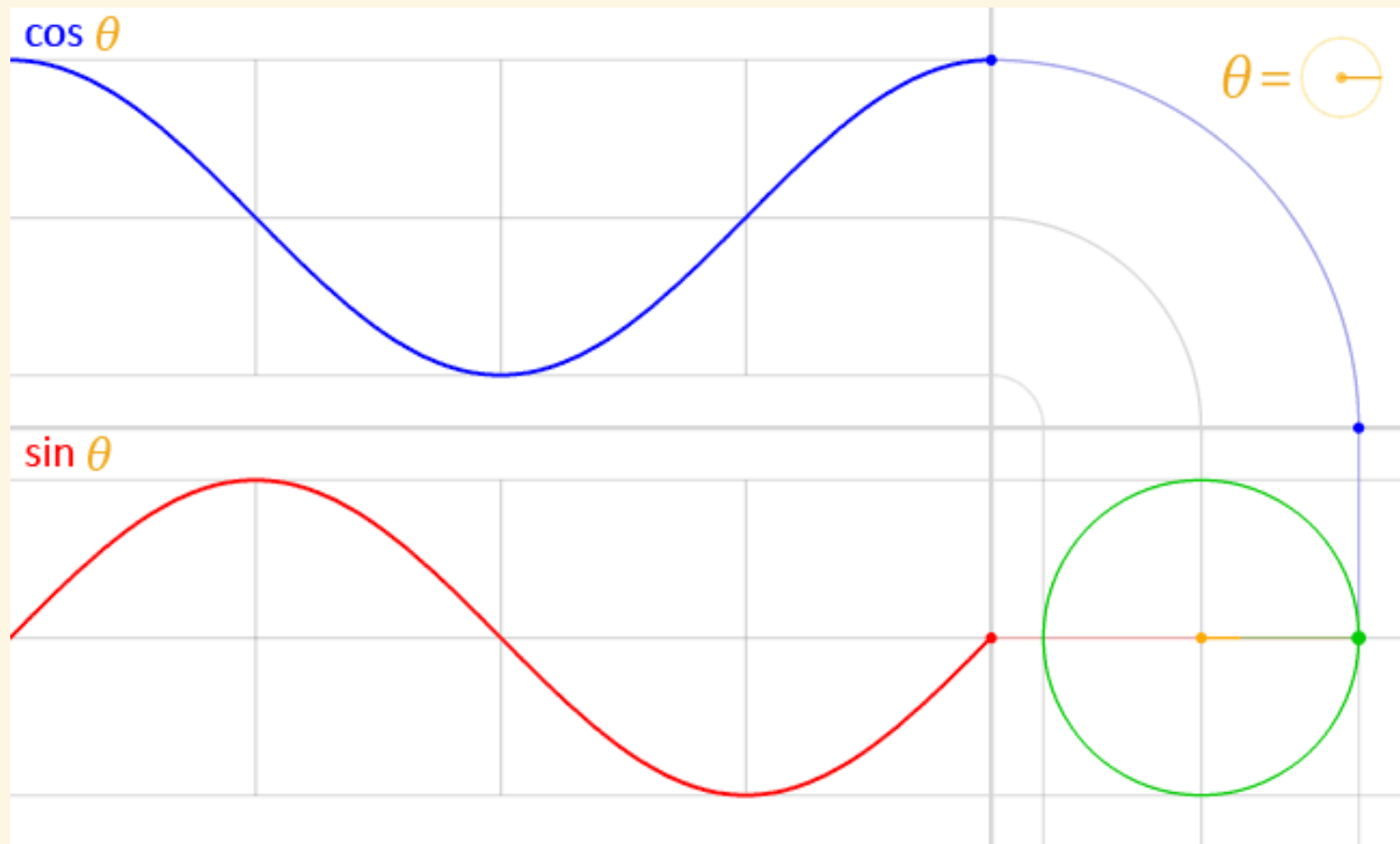
$$y = \sin(x)$$



Trigonometric functions



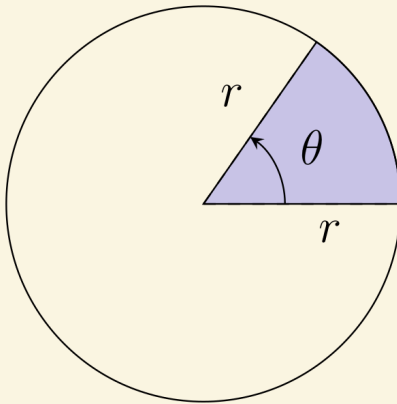
Trigonometric functions



Degrees vs Radians

$$360 \text{ degrees} = 2\pi \text{ radians}$$

where $\pi = 3.14159265359 \dots$



Degrees:

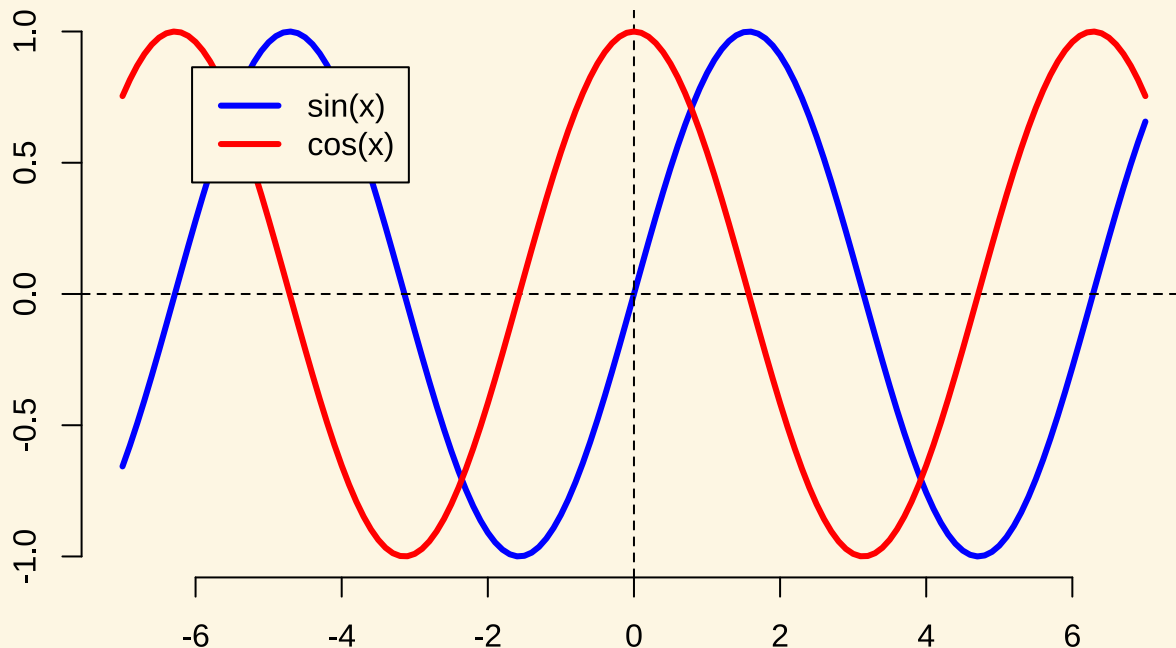
$$\text{Area} = \pi r^2 \times \frac{\theta}{360}$$

Radians:

$$\text{Area} = \pi r^2 \times \frac{\theta}{2\pi} = \frac{1}{2}\theta r^2$$

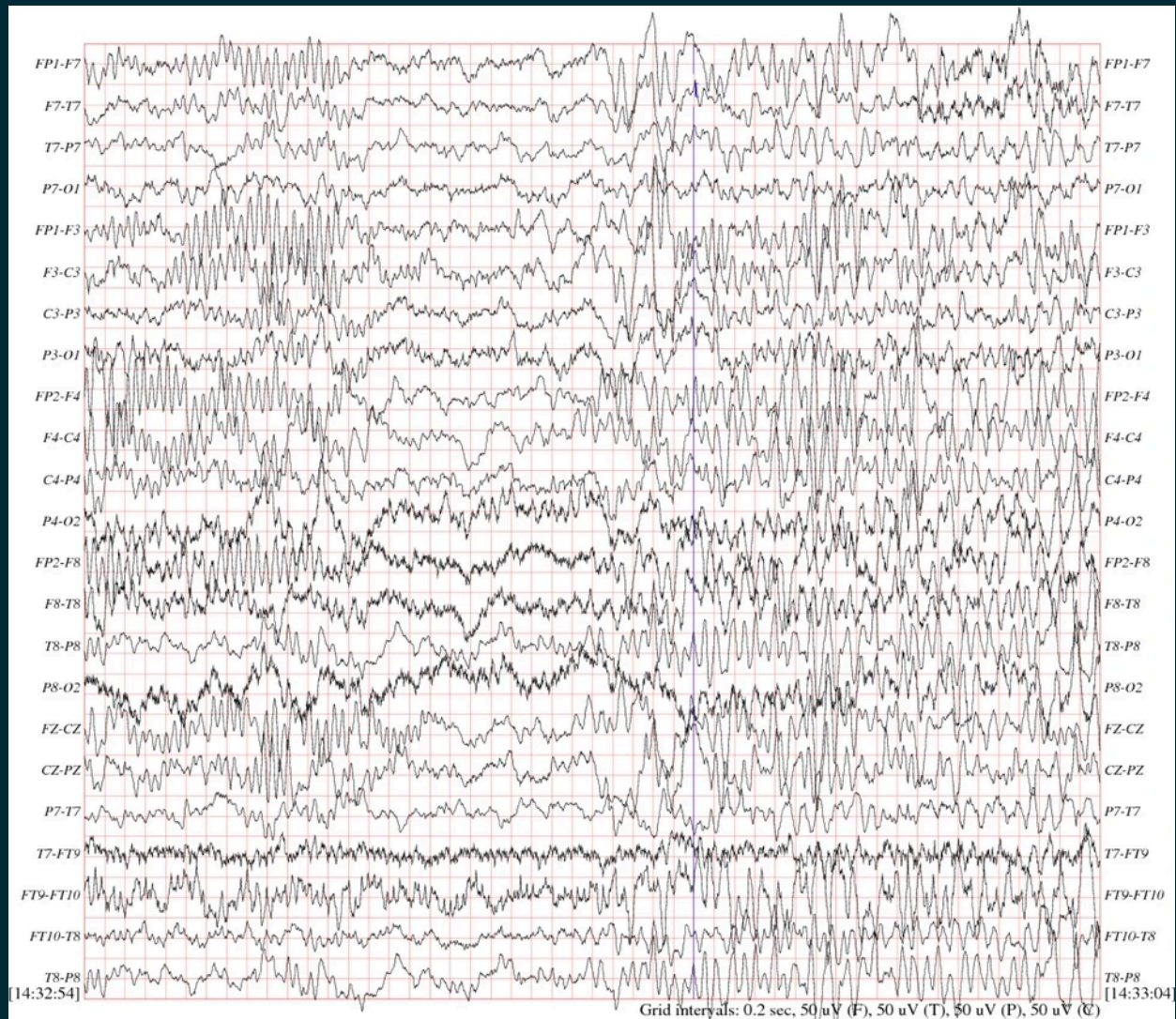
Usually radians make calculations easier.

Trigonometric functions

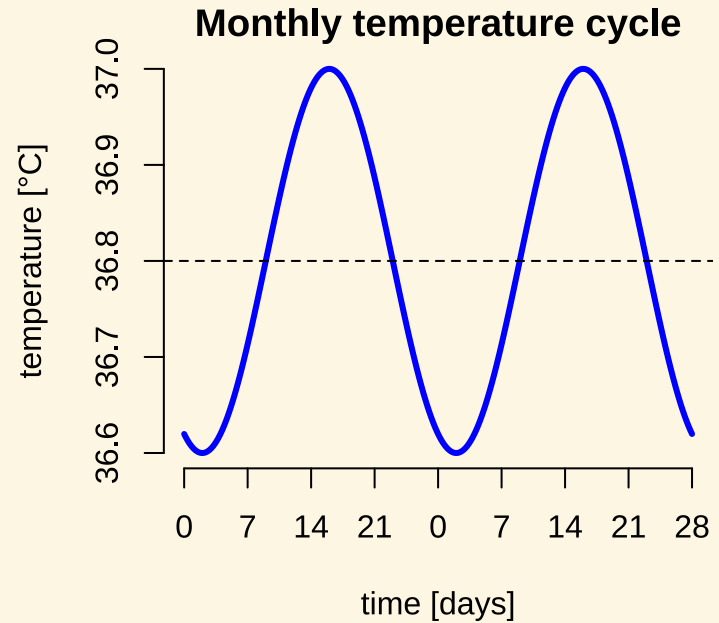
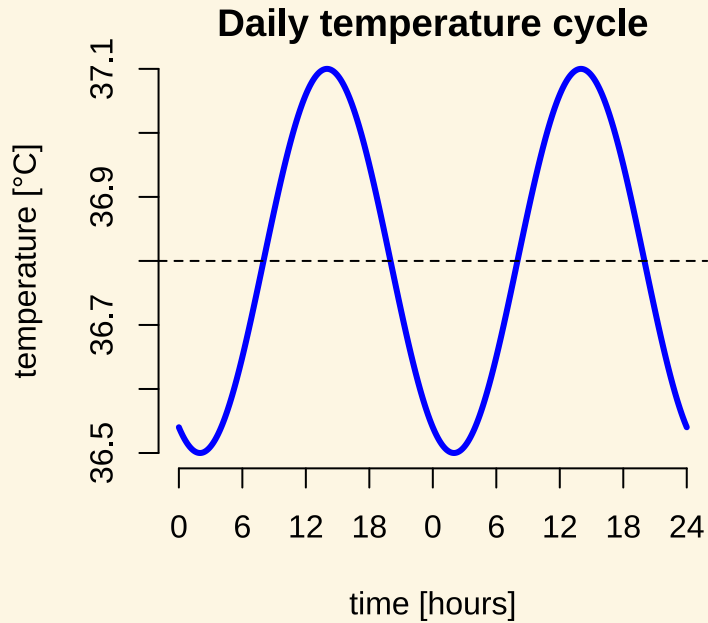


- Sine and cosine functions take angles as input, and return numbers between -1 and 1 as outputs
- Can be used to model *oscillations* (phenomena with repeated/period variation)
- Real-world oscillations are not simply periodical

Oscillations in scalp EEG activity



Monthly cycle in women's body temperature



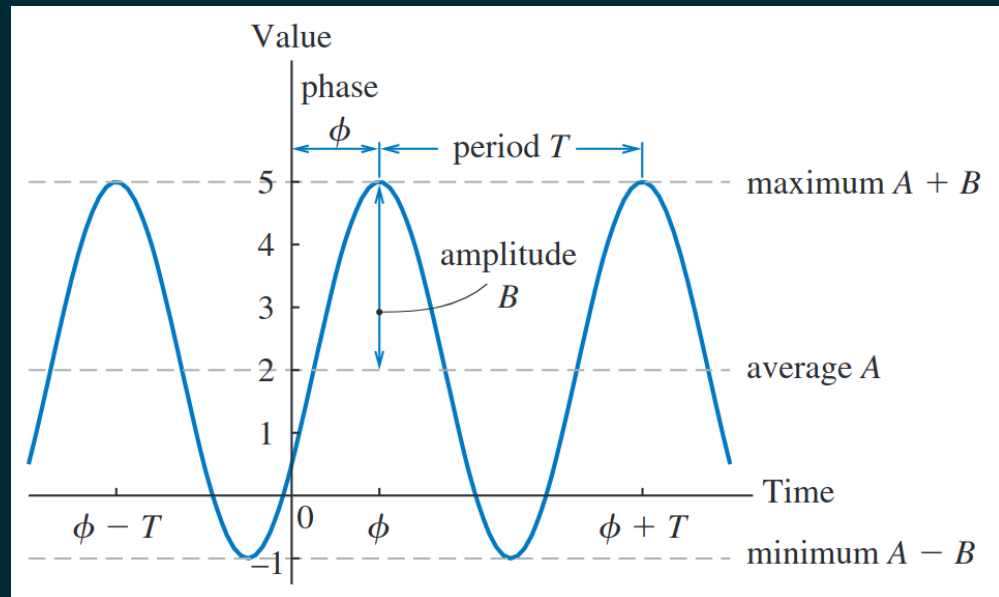
$$D(t) = 36.8 + 0.3 \cos\left(\frac{2\pi}{24}(t - 14)\right)$$

$$M(t) = 36.8 + 0.2 \cos\left(\frac{2\pi}{28}(t - 16)\right)$$

Modelling oscillations with cosine function

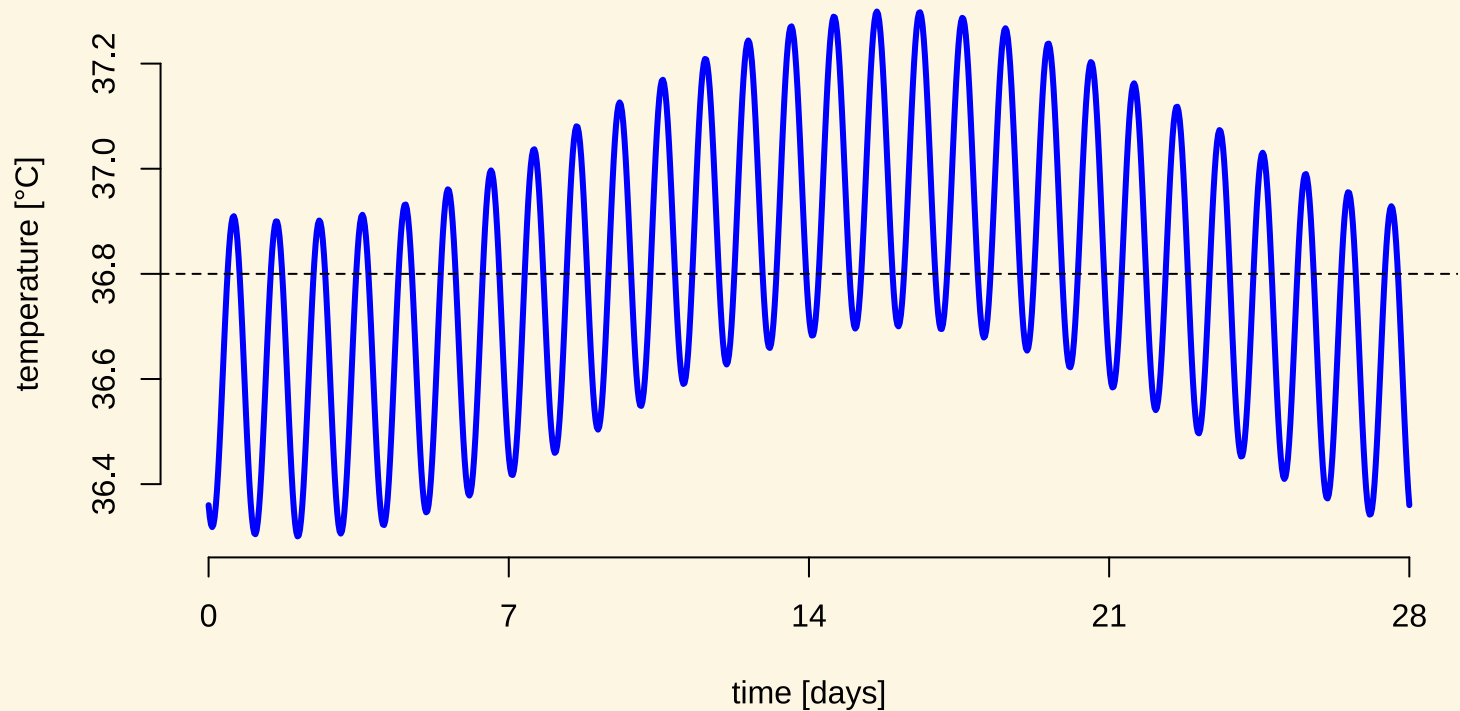
$$y(t) = A + B \cos\left(\frac{2\pi}{T}(t - \phi)\right)$$

- A , average
- B , amplitude
- T , period
- ϕ , phase (the time of the first peak)



Additionally, we can describe an oscillation according to its frequency (the inverse of the period), $f = \frac{1}{T}$. Frequency is often measured in 'hertz' (number of cycles per second).

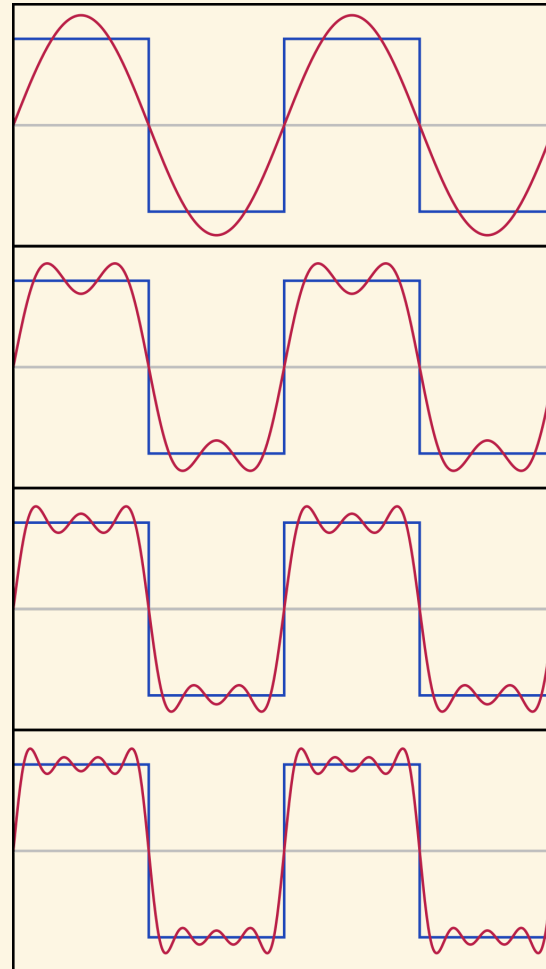
Monthly cycle in women's body temperature



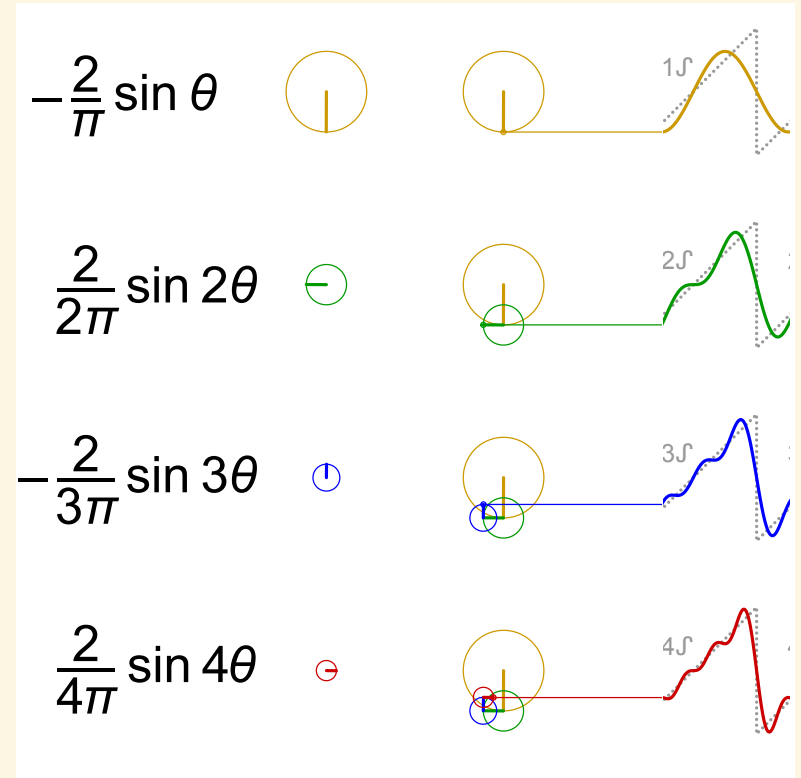
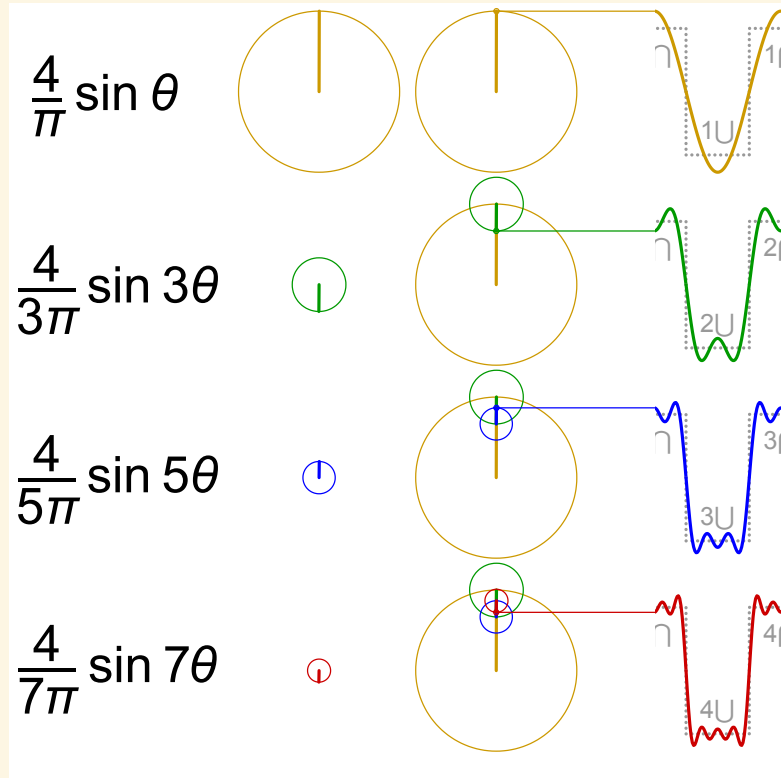
$$f(t) = 36.8 + \underbrace{0.3 \cos\left(\frac{2\pi}{1}\left(t - \frac{14}{24}\right)\right)}_{\substack{\text{daily oscillation} \\ \text{(period } T \text{ and phase } \phi \\ \text{transformed in days)}}} + \underbrace{0.2 \cos\left(\frac{2\pi}{28}(t - 16)\right)}_{\text{monthly oscillation}}$$

Fourier series

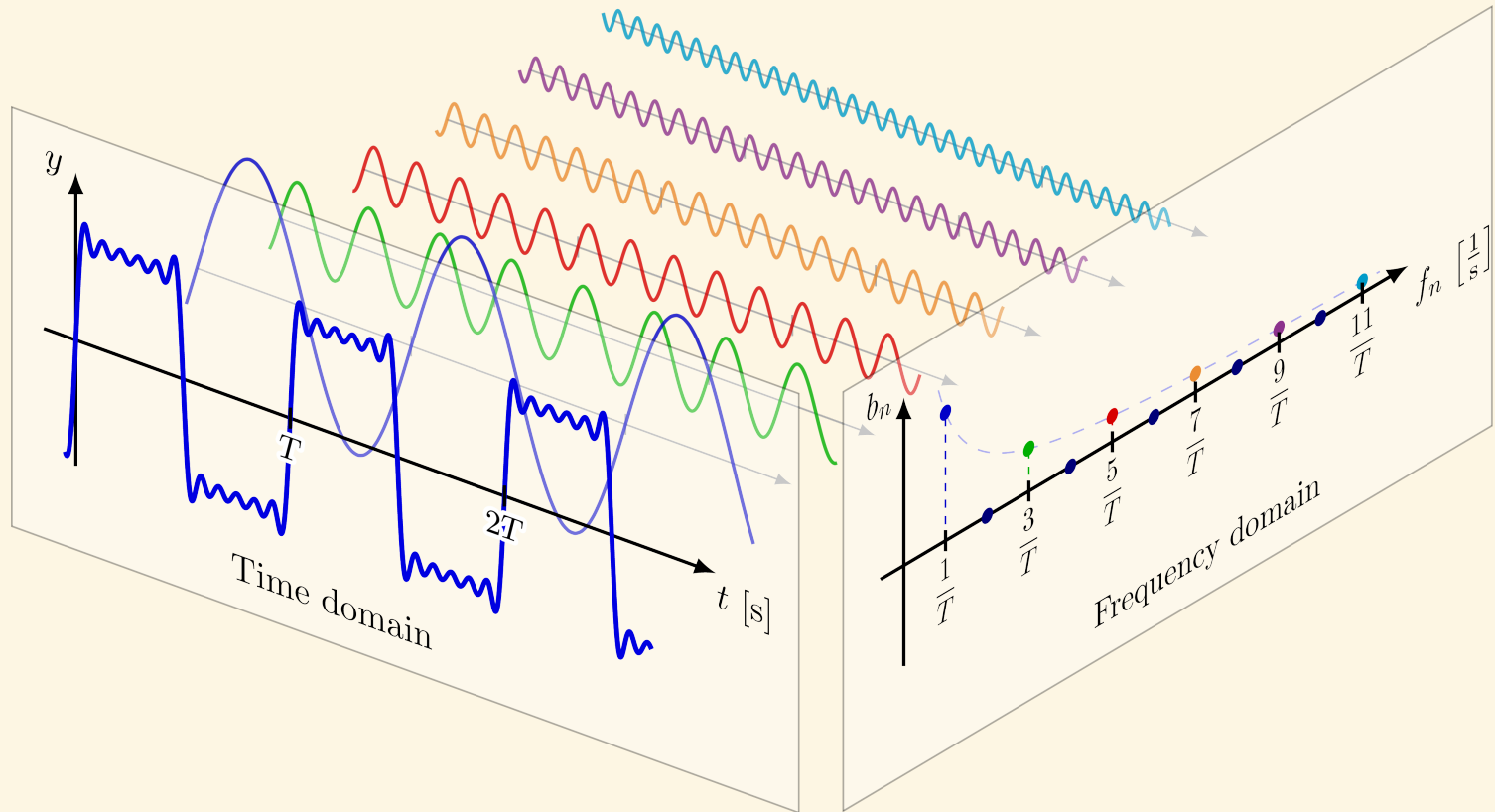
- We can approximate almost any functions by summing many cosine functions with different amplitudes, periods and phases.
- The approximation improves as the number of terms (sinusoidal functions added together) increases



Fourier series



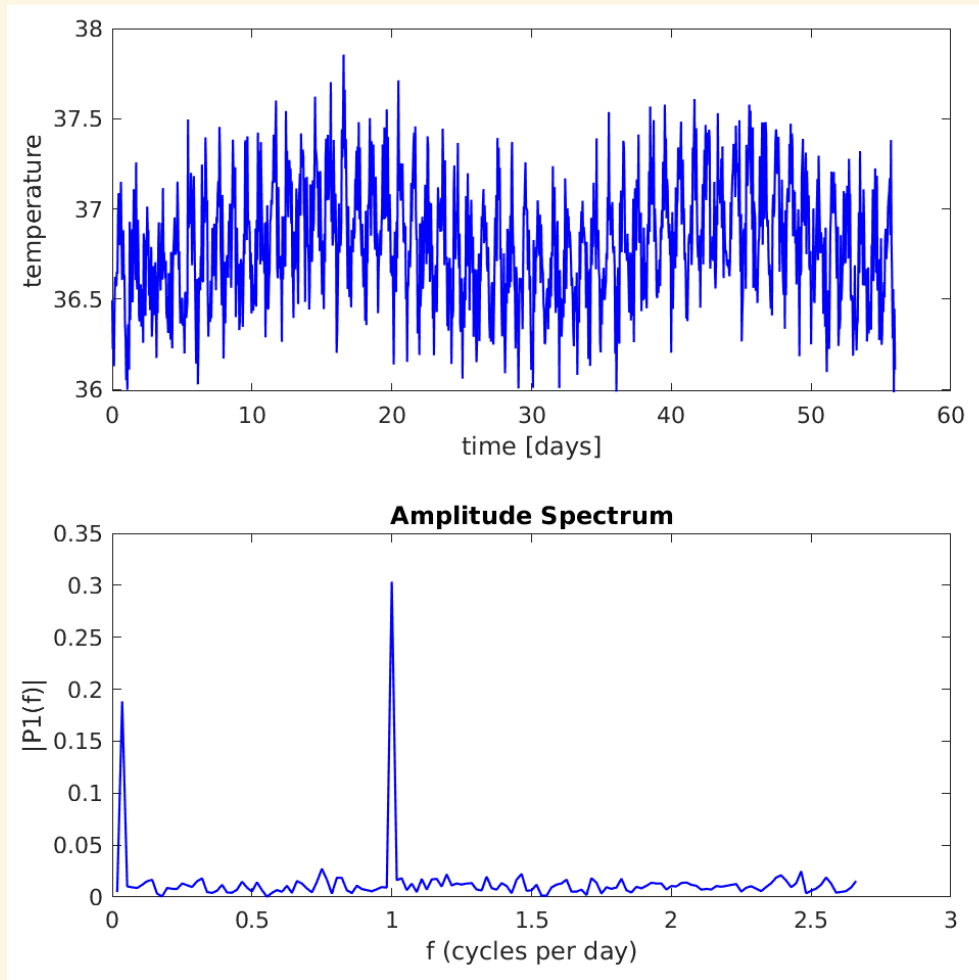
Fourier transform



Fourier transform is a powerful mathematical tool that decompose a function into different frequency components. In the case of a signal that varies over time this corresponding to moving from a representation in the time domain to a frequency domain.

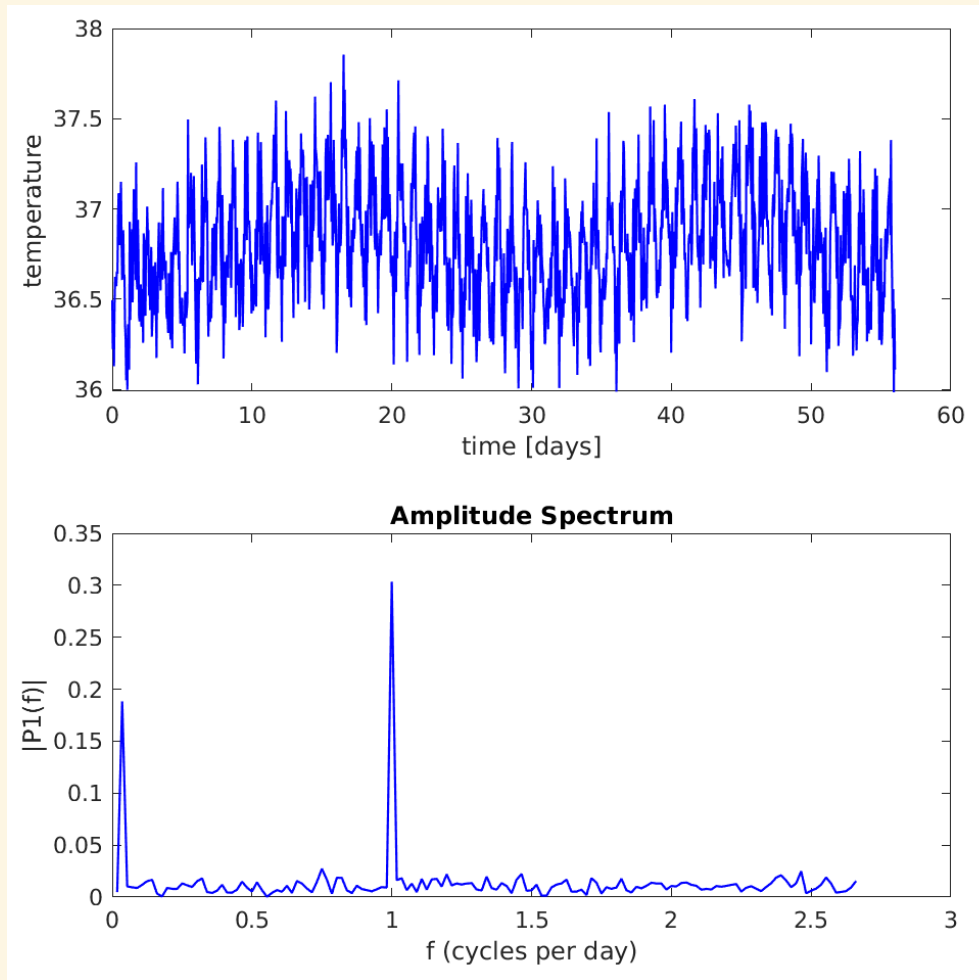
Fourier transform & filtering

- Frequency-domain representation of body temperature signal.
- (hourly measurement, random noise added)



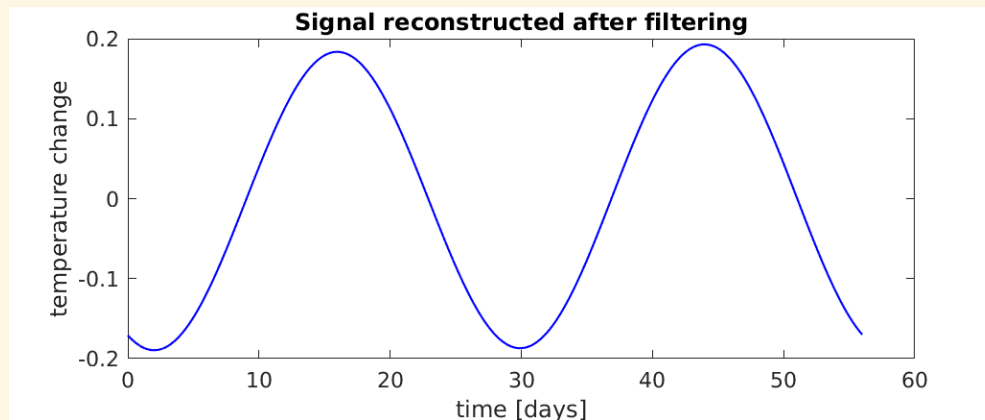
Fourier transform & filtering

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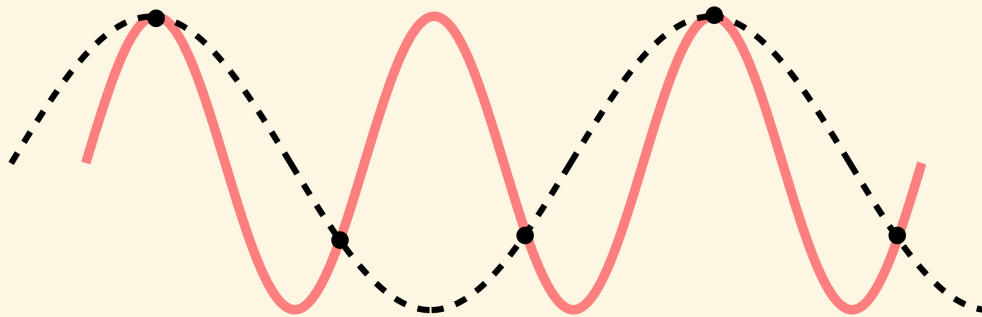
Fourier transform & filtering

- Working on the frequency-domain representation makes it easy to suppress frequencies selectively
- Low pass filter (I set to zero everything with a period shorter than 2 days)



Continuous & discrete signals

- Continuous functions can be used to describe *continuous* (analog) signals (e.g. voltage of a EEG electrode)
- In practice, most of the time we work with *discrete* signals - that is sequence of values. that comes from a sampling (or discretization) of a continuous signal. For example:
 - recording of EEG voltage is stored as a sequence of values;
 - in eyetracking, movements of visual axis are discretized in a sequence of gaze positions.
- We can apply similar operations to both types of signals (with some caveat). For example we have the *discrete Fourier transform* for discrete signals.
- Information is lost in the sampling process. If the sampling interval (distance between sampling points) is too coarse, then the continuous signal is not uniquely determined by the samples causing *aliasing* errors.



Aliasing error in the wild



Application of Fourier transform in two dimensions



Part 1 summary

Functions map inputs to outputs and express relationships between variables (continuous or discrete).

Differentiation captures local rate of change—slopes and tangents; on sampled data we approximate it with finite differences.

Exponentials & logs are opposite operations: they turn products into sums.

Probability functions describe uncertainty: densities (continuous, e.g. Normal distribution) vs masses (discrete, e.g. binomial).

Trigonometric functions (sine/cosine) are useful to model oscillatory signals;

Fourier transform moves between time and frequency representations of a signal and enable filtering of signals.